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Musculoskeletal parameters of muscles crossing the shoulder and elbow and the effect of sarcomere length sample size on estimation of optimal muscle length

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Abstract

Background. Knowledge of musculoskeletal parameters is essential to understanding and modeling a muscle's force generating capability. A study of musculoskeletal parameters was conducted in two parts: (I) Empirical measurement of upper extremity musculoskeletal parameters. (II) Computational bootstrap simulation to examine statistical power of detecting optimal muscle length as a function of sarcomere length sample size and effect size.

Methods. Parameters were determined with a cadaver model. Sarcomere lengths were measured for 120 samples per muscle using laser diffraction and the mean sarcomere length used to estimate optimal muscle length. A bootstrap computational simulation was conducted to estimate variance in mean sarcomere length as a function of sample size. Statistical power for detecting optimal muscle length as a function of sample size and effect size was then determined.

Findings. Parameters are reported in tabular format. Power is 80% at approximately 85, 50, 40 and 25 samples for effect sizes of 0.5, 0.75, 1.0 and 1.5 mm respectively.

Interpretation. Musculoskeletal parameters for predicting muscle forces can be adequately measured in a cadaver model. Measurement of 40–60 sarcomere lengths per muscle is sufficient to calculate mean sarcomere length for estimating optimal muscle length with power of 80% for an effect size of 0.75–1.0 mm.
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1. Introduction

Physiologic cross-sectional area (PCSA) affects the maximum force a muscle can produce, and optimal muscle length (OML), tendon cross-sectional area and compliance affect where a muscle acts on its length tension curve (Zajac, 1989). Tendon compliance can be modeled by equating muscle and tendon force, resulting in a non-linear constraint that allows muscle and tendon lengths to be determined for a given muscle force. However, applying the constraint requires knowledge of tendon cross-sectional area and tendon length.

Although each individual parameter is important, the muscle force depends on a set of multiple parameters. Consequently, it would be ideal to develop models using parameters measured from the same specimens (Murray et al., 2000).

Previous studies (Amis et al., 1979; An et al., 1981; Bassett et al., 1990; Van der Helm and Veenbaas, 1991; Veeger et al., 1991, 1997) document PCSA and fascicle lengths of upper extremity muscles. Other studies estimate OML for upper extremity muscles (Klein Breteler et al., 1999; Murray et al., 2000). However, no studies determined PCSA and OML in conjunction with tendon

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slack length and tendon cross-sectional area for all muscles of the glenohumeral joint. Thus, there is a need for a comprehensive dataset suitable for predicting forces of muscles spanning the glenohumeral joint.

Previous studies (Murray et al., 2000; Delp et al., 2001) have estimated OML from measurements made for muscle sarcomere lengths. The approach has been to make a relatively large number of sarcomere measurements (120) from various regions of the muscle to calculate a mean sarcomere length and scale mean sarcomere length by optimal sarcomere length (2.8 μm) to estimate muscle optimal length. Given the implied assumptions of utilizing the mean sarcomere length to estimate OML, there has been little attention given to the effects of sarcomere length sample size on variance of mean sarcomere length and variance of OML, and the effects of this variance on statistical power to detect an OML estimated using a full dataset of 120 sarcomere lengths.

The purpose of this study was to empirically determine a complete dataset of musculoskeletal parameters, including PCSA, OML, tendon area, and tendon slack length for muscles crossing the glenohumeral joint, and elbow flexors and extensors. Secondly, using bootstrap simulation, we wanted to determine the effects of sarcomere sample size on variance of mean sarcomere length, and subsequently estimate statistical power to detect a ''gold standard'' OML.

2. Methods

2.1. Empirical cadaver measurements

Musculoskeletal data was collected from two unembalmed human female cadaver upper extremities obtained from anatomical donations. The ages and weights at time of death were 28 years, 65.9 kg and 91 years, 63.6 kg, and both specimens were 1.65 m in height. Prior to harvesting the upper extremities, anthropometric measurements were made on the cadavers (Table 1). Care was taken to harvest with latisimus dorsi and pectoralis major completely intact. Upper extremities were frozen at −20 °C for a period of approximately two weeks. Twenty-four hours prior to dissection upper extremities were thawed in a room temperature water bath. Care was taken when dissecting to remove all muscle and tendon from origins or insertions. Muscles with large insertion sites, or large variations in pennation angle and/or muscle fascicle length were dissected into sub-regions as shown in Table 2. The literature was consulted (Hogfors et al., 1987; Van der Helm and Veenbaas, 1991) to determine the best divisions of the muscles into sub-regions. Each muscle sub-region was treated as an individual muscle for purposes of parameter measurement. Muscles that were not dissected were covered with saline soaked gauze to retard dehydration. Parameters were sequentially measured on each musculotendinous unit immediately following dissection. When possible, a muscle was manipulated into a parallelpipidon shape (An et al., 1981; Brand et al., 1981) for measurement of muscle belly length, muscle fascicle length, tendon and aponeurosis length, and pennation angle. If a muscle could not be manipulated into the parallelpipidon shape, it was positioned in a manner consistent with anatomy. The musculotendinous unit was placed flat upon a table and tendon length and aponeurosis lengths were measured. Tendon was dissected from the muscle and tendon volume was determined by water displacement. Pennation angle, the angle between the muscle fascicles and a line connecting distal and proximal tendons, was measured with a goniometer to the nearest degree. Muscle belly length and average muscle fascicle length were measured with a digital caliper to the nearest millimeter, and muscle volume was measured via water immersion. Muscles were placed in a solution of 20% neutral buffered formalin (NBF) for a period of 7–9 days for fixation.

Following fixation, thin slips of fascicles were dissected from the muscle interior, below the outer fascicle layer, using fine sharp dissection. To ensure fascicles were dissected from varying regions, muscles were visually divided into proximal, middle and distal regions, and three, four, and three fascicles were dissected from each proximal, middle and distal region, respectively. Fascicles were mounted on microscope slides using Kaiser's jelly and cover slipped. Sarcomere lengths were determined using laser diffraction (Lieber et al., 1990; Murray et al., 2000; Delp et al., 2001) with a custom built diffractometer. The width of the first order diffraction band was measured and used to determine the sarcomere length, l^s:

$$l^s = \frac{\lambda}{\sin\theta} \quad \text{and} \quad \tan\theta = \frac{y}{L}, \tag{1}$$

Table 1
Cadaver anthropometry

Specimen	Chest A-P depth (cm)	Thorax length (cm)	Upper arm circumference (cm)	Palpable humerus length (cm)	Carrying angle pre-dissection (deg)	Skinned carrying angle (deg)
1	23.8	48.3	26.0	27.6	15	12
2	17.1	50.8	31.8	29.8	9	11

Table 2
Summarized musculoskeletal parameters (mean and standard error of the mean)

	Tendon length (mm)	Tendon area (cm^2)	PCSA (cm^2)	Pennation angle (deg)	Muscle length (mm)	Fascicle length (mm)	Mean sarcomere length (μm)	Optimal fascicle length (mm)	Optimal muscle length (mm)
Bicep (LH)	182.8 (13.8)	0.161 (0.043)	1.57 (0.13)	0 (0)	162.5 (15.5)	145.5 (9.5)	2.61 (0.12)	156.1 (3.0)	174.1 (8.6)
Bicep (SH)	157.5 (8.5)	0.191 (0.010)	1.75 (0.79)	0 (0)	183.5 (20.5)	160.0 (12.0)	2.47 (0.13)	180.9 (3.8)	207.0 (12.0)
Tric. (Lat.)	167.0 (6.5)	0.240 (0.009)	4.13 (1.22)	26 (8.0)	144.0 (10.0)	82.5 (14.5)	2.28 (0.15)	102.8 (24.4)	176.6 (0.9)
Tric. (Long)	199.5 (6.0)	0.272 (0.117)	3.60 (1.35)	12 (6.5)	208.0 (39.0)	136.5 (1.5)	2.18 (0.11)	176.2 (10.5)	265.8 (37.2)
Tric. (Med.)	17.8 (11.3)	0 (0)	3.21 (0.71)	17 (1.0)	178.5 (19.5)	120.0 (9.0)	2.32 (0.04)	144.6 (8.7)	215.0 (20.3)
Ant. Delt.	26.0 (23.0)	0.102 (0.102)	5.46 (2.41)	22 (5.0)	126.0 (20.0)	98.5 (16.5)	2.71 (0.38)	101.2 (3.0)	129.6 (2.7)
Mid. Delt.	15.5 (15.5)	0 (0)	7.39 (3.32)	15 (6.5)	135.5 (7.5)	116.5 (2.5)	2.93 (0.23)	111.8 (6.2)	129.7 (2.8)
Pos. Delt.	40.0 (8.0)	0.104 (0.104)	4.69 (0.02)	18 (2.0)	153.0 (9.0)	119.5 (5.5)	2.42 (0.32)	141.8 (25.2)	181.8 (34.6)
Coracobrach.	17.0 (17.0)	0 (0)	1.67 (0.55)	27 (5.5)	131.5 (5.5)	78.0 (10.0)	2.30 (0.19)	96.7 (20.2)	162.0 (20.1)
Pec. Maj. (Clav.)	22.8 (6.8)	0.651 (0.026)	3.07 (0.70)	17 (1.0)	154.0 (25.0)	137.0 (18.0)	2.61 (0.20)	149.5 (31.0)	168.4 (40.0)
Pec. Maj. (Ster.)	46.8 (8.3)	0.662 (0.117)	5.68 (0.12)	25 (5.0)	171.0 (7.0)	143.0 (8.0)	2.79 (0.11)	143.6 (2.2)	171.8 (0.1)
Lat. (Sup.)	83.3 (4.8)	0.425 (0.084)	2.11 (0.31)	25 (2.5)	227.0 (18.0)	183.5 (35.5)	1.93 (0.15)	263.4 (31.2)	328.7 (0.9)
Lat. (Mid.)	96.5 (21.0)	0.236 (0.104)	2.58 (0.40)	19 (2.5)	283.0 (6.0)	185.0 (20.0)	2.14 (0.09)	241.0 (16.4)	369.9 (7.1)
Lat. (Inf.)	80.3 (13.8)	0.235 (0.084)	2.61 (0.34)	21 (7.0)	316.0 (26.0)	243.5 (14.5)	2.36 (0.04)	289.2 (12.5)	375.2 (24.8)
Supraspinatus	29.5 (6.5)	0.495 (0.060)	3.36 (0.52)	7 (7.0)	85.0 (4.0)	74.0 (0)	2.94 (0.17)	70.7 (4.0)	81.4 (8.4)
Teres minor	23.3 (8.3)	0.492 (0.175)	2.44 (1.03)	24 (8.5)	115.5 (17.5)	66.0 (7.0)	2.54 (0.48)	76.8 (22.1)	128.1 (4.7)
Infra. (Sup.)	50.8 (1.3)	0.298 (0.106)	3.78 (1.63)	16 (2.5)	117.5 (1.5)	83.0 (10.0)	2.79 (0.46)	87.4 (24.6)	121.5 (21.7)
Infras. (Inf.)	50.8 (2.8)	0.198 (0.011)	4.56 (0.49)	21 (5.5)	98.0 (31.0)	73.5 (20.5)	2.92 (0.40)	69.1 (10.1)	91.6 (17.0)
Subscap. (Sup.)	39.8 (1.8)	0.373 (0.109)	2.26 (0.83)	19 (2.5)	85.5 (1.5)	66.0 (5.0)	2.16 (0.20)	85.8 (1.5)	112.2 (12.4)
Subscap. (Mid.)	38.8 (4.3)	0.261 (0.029)	4.09 (0.50)	24 (3.0)	102.5 (9.5)	70.0 (2.0)	2.18 (0.17)	90.5 (9.7)	133.2 (22.6)
Subscap. (Inf.)	28.3 (7.3)	0.379 (0.097)	3.14 (0.47)	17 (1.0)	100.0 (15.0)	71.5 (9.5)	2.10 (0)	95.3 (12.8)	133.3 (20.2)
Teres major	25.5 (16.0)	0.120 (0.120)	2.93 (0.28)	16 (9.5)	140.0 (11.0)	123.5 (7.5)	2.06 (0.03)	168.4 (12.9)	191.0 (18.0)
Brachialis	33.5 (4.5)	0.132 (0.132)	7.71 (3.28)	18 (6.0)	151.0 (15.0)	87.5 (15.5)	2.65 (0.19)	94.2 (23.2)	159.4 (4.4)
Brachioradialis	103.5 (5.0)	0.148 (0.055)	1.15 (0.37)	0 (0)	167.0 (7.0)	155.5 (2.5)	2.51 (0.22)	175.3 (17.9)	188.7 (24.1)

where L is the distance from the fascicle to the defraction pattern, and y is the width of the first order band. The wavelength, λ, of the 5 mW helium–neon laser was 632.8 nm.

Twelve sarcomere measurements were made on each fascicle for a total of 120 sarcomere lengths per muscle or muscle sub-region (Murray et al., 2000; Delp et al., 2001). In very rare circumstances no diffraction pattern was observable. This problem was attributed to specimens consisting of non-contractile proteins. Due to the diffuse diffraction pattern that can be generated by the muscle tissue, a pilot study was conducted to assess the interrater reliability (Portney and Watkins, 1993) of sarcomere measurements. Two people independently measured the diffraction pattern of 40 muscle specimens.

Optimal fascicle lengths, l_o^f were determined by normalizing measured fascicle length, l^f, by the ratio of optimal sarcomere length of 2.8 μm (Walker and Schrodt, 1974) to the measured sarcomere length:

$$l_o^f = l^f \frac{2.8}{l^s}. \tag{2}$$

OML was estimated as measured muscle length normalized to an optimal sarcomere length of 2.8 μm.

Tendon length, L^T, was determined by the following formula (Cheng and Scott, 2000), in which one half the aponeurosis length at origin, L_{org}^A, and insertion, L_{ins}^A, was added to tendon lengths measured at the origin, L_{org}^T, and insertion, L_{ins}^T:

$$L^T = 0.5(L_{org}^A + L_{ins}^A) + (L_{org}^T + L_{ins}^T). \tag{3}$$

Effective tendon cross-sectional area was determined as the ratio of tendon volume to tendon length. Likewise, muscle PCSA was determined as muscle volume divided by optimal fascicle length.

2.2. Bootstrap computational simulation

Bootstrap simulation can be used to non-parametrically examine variance in empirical data as well as to determine confidence intervals and to simulate the effects of sample size on variability (Efron, 1979; Mooney and Duval, 1993). Bootstrap methods have been used by others to estimate the relationship between power and sample size for a given experimental paradigm (Mathiassen et al., 2002; Van Dieen et al., 2002).

The purpose of the bootstrap simulation was to examine how measuring less than 120 sarcomeres effects the variance of the mean sarcomere length, the variance of the predicted OML and the power to detect the predicted OML determined using the full 120 sarcomere length sample as a "gold standard". The data input to the simulation consisted of all 120 sarcomere lengths measured for one muscle, Short Head Biceps, of one specimen reported in Table 2. The data was resampled "n" times (where n represents the simulated number of measured sarcomere lengths, 110–10 samples in 10 sample increments), with replacement, and mean sarcomere length was determined at the resample level "n". 1000 resamples were taken at each resample level to establish distributions of the mean sarcomere length. That is, at a given resample level "n", the bootstrap model simulates an experiment conducted according to the protocol outlined earlier measuring "n" sarcomere lengths. The variance in mean sarcomere length was then calculated from the distributions, and used to calculate the power for detecting the "gold standard" optimal muscle length. Specifically, the statistical power of the Welch test of equality of means (Welch, 1951) was determined after correcting for unequal sample size and unequal variance among groups (Kulinskaya et al., 2003) with probability of type I error (α) set equal to 0.05.

3. Results

3.1. Empirical cadaver measurements

The intraclass correlation for sarcomere lengths measured by two people was 0.99. Mean and standard error of the mean for the measured parameters are reported in Table 2. Mean lengths of sarcomeres from biceps, triceps, posterior deltoid, clavicular head of pectoralis major, coracobrachialis, latisimus dorsi, teres minor, subscapularis, teres major, brachialis and brachioradialis were less than the optimal sarcomere length of 2.8 μm. Consequently, for these muscles estimated OML was greater than measured muscle belly length. Mean lengths of sarcomeres from anterior deltoid, middle deltoid, sternal head of pectoralis major, supraspinatus, and infraspinatus were near 2.8 μm, and in these muscles estimated OML was near measured muscle belly length. Interestingly, no sarcomere lengths were found to be excessively greater than the optimal sarcomere length.

The muscles with largest PCSA were brachialis (7.71 cm^2), sternal head of pectoralis major (5.68 cm^2), and anterior deltoid (5.46 cm^2). Brachioradialis (1.15 cm^2), long head of biceps (1.57 cm^2), and coracobrachialis (1.67 cm^2) possessed the smallest PCSA.

Tendon cross-sectional areas for medial head of triceps, middle deltoid and coracobrachialis could not be determined due to a lack of tendon external to the muscle measurable via volume displacement. Anterior (0.102 cm^2) and posterior deltoid (0.104 cm^2) had the smallest tendon cross-sectional areas, while sternal head of pectoralis major (0.662 cm^2), clavicular head of pectoralis major (0.651 cm^2), and supraspinatus (0.495 cm^2) had the largest.

Muscles with the longest tendon lengths; long (182.8 mm) and short head of bicep (157.5 mm), lateral (167.0

mm) and long head of tricep (199.5 mm) and brachioradialis (103.5 mm), were fusiform, of negligible pennation, and are expected to undergo the greatest excursion through a range of joint motion. Tendon lengths for medial head of triceps (17.8 mm), middle deltoid (15.5 mm) and coracobrachialis (17.0 mm) were short due to a lack of tendon external to the muscle belly in spite of significant aponeurosis.

3.2. Computational bootstrap simulation

Standard deviation of predicted OML as a function of the number of sarcomere length samples is shown in Fig. 1. Standard deviation for a resample level of 120 is of order of 1.25 mm and increases to approximately 4 mm for resample level of 10. Power to detect the OML is shown as a function of sarcomere samples for various effect sizes in Fig. 2. For effect sizes of 0.75 mm and greater, power decreases from 99% at 110 samples and is 80% at approximately 50, 40 and 25 samples for effect sizes of 0.75, 1.0 and 1.5 mm respectively. For effect size of 0.5 mm, power is initially greater than 90% for 110 samples and decreases to 80% at approximately 85 samples.

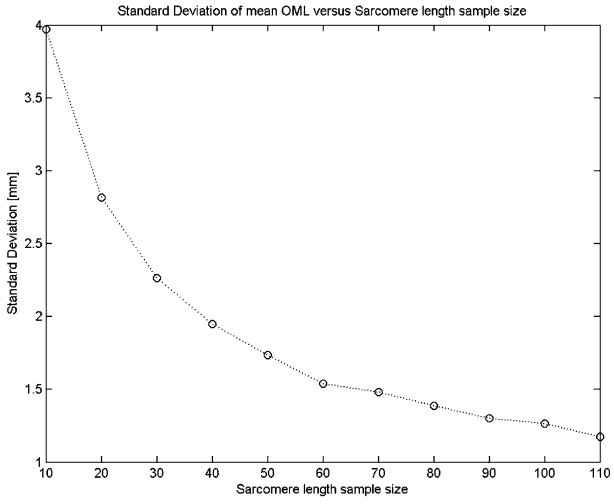


Fig. 1. Standard deviation of mean optimal muscle length versus sarcomere length sample size.

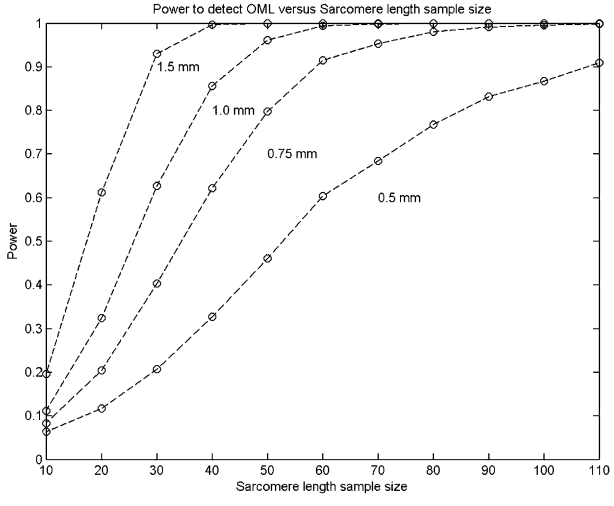


Fig. 2. Power to detect optimal muscle length versus sarcomere length sample size by effect size.

4. Discussion

This study generated data for use in developing computational models of the musculature crossing the glenohumeral and elbow joints. Most importantly, it provides estimates of the optimal muscle lengths, which are critical parameters in many models. In addition, the data supplied here can be used to design sarcomere length sampling strategies for future studies.

Most biomechanical models incorporating the force–length property of muscle require information relating muscle length as a function of joint angle to the muscle's optimal length (Kaufman et al., 1989; Zajac, 1989) For simplicity these models use data from previous studies to describe this relationship. However it is important to understand sources of variability in the data, and how the variability affects predicted muscle force. Only then can modelers understand the implications of using optimal muscle length parameters based on one subject versus OMLs based on a group of subjects. In common mathematical formulations of the force–length property of muscle, a strain factor is determined by normalizing the muscle length against the optimal muscle length. To compare the sources of variability we will consider the measured and simulated standard deviations for OML and also effect sizes as "error measures", and calculate muscle strain errors by normalizing each error measure against the mean OML. Ideally we could express the "error measures" not as muscle strain errors but as muscle force errors, however since the length–tension relationship for most upper extremity muscles is unknown, the determination of muscle force errors is an intractable problem.

It is common to observe between specimen standard deviation of the mean OML of 4.2 cm magnitude (Table 2 this study, and Murray et al., 2000, Table 2). This standard deviation for OML translates to muscle strain error of 18%. The bootstrap estimated standard deviation for the short head of biceps for one muscle of one specimen in this study (Fig. 1) varied from 1.25 to 4.0 mm, translating to muscle strain error of 0.6–2.1%. At the specimen level, an effect size of 1.0 mm translates to a muscle strain error of 0.5%.

Looking at variability in this manner one can see that more variability exists between specimens than within a specimen. The simulation suggests it is sufficient to

measure from 40 to 60 sarcomere lengths per specimen and use resulting mean sarcomere length to estimate OML. The 40–60 sarcomere lengths yields power of approximately 80% for an effect size of 0.75–1.0 mm which is smaller than the between specimen standard deviation of the mean for most muscles.

We provide data for planning studies involving sarcomere measurements and using these measurements to estimate optimal muscle length. The tools presented here will assist experimentalists in designing experiments to estimate mean sarcomere length, and subsequently optimal muscle length with a desired effect size and statistical power. While the effect of between subject variability is interesting and warrants investigation, it is important to note that in this study we did not examine the effect of the number of specimens, i.e. between specimen variability, on estimation of optimal muscle length, but rather the effect of the number of sarcomere measurements for a given specimen, i.e. within specimen variability, on optimal muscle length.

The most serious limitation of this study was the use of cadaver tissue for estimating optimal muscle lengths. Although in vivo measurement of sarcomere length using laser diffractometry is ideal (Lieber and Friden, 1997, 1998), the exposure necessary to achieve that for the shoulder is unrealistic. Thus, cadaver measurements are the most feasible method available (Lieber et al., 1990; Delp et al., 2001; Murray et al., 2000). An additional limitation was that not all muscles of the torso and upper extremity were measured. Although this limits the utility of the data to musculoskeletal modelers interested in the neck and torso, the study provides important data on muscles crossing the glenohumeral joint. Given, the clinical importance of the rotator cuff, this is an important contribution for modeling common shoulder pathology.

A strength of the study was the careful measurement of sarcomere length using laser diffractometry. The results of the pilot study determined an intraclass correlation coefficient of 0.99, indicating excellent reliability. Another strength is that PCSA and OML were determined in conjunction with tendon slack length and tendon cross-sectional area for all muscles.

One shortcoming of this study is that optimal muscle length alone is not sufficient for predicting muscle force throughout a range of motion. An accurate description of muscle strength requires information relating muscle length to joint angle. Additionally, since muscle force cannot be measured directly for many muscles of the upper extremity, but joint torque can be measured, one additional piece of information is necessary, muscle moment arm. By knowing muscle moment arm as a function of joint angle a muscle's contribution to total joint torque can be evaluated. For the upper extremity, a perfect understanding of these relationships has been elusive. Only when muscle length and moment arm as a function of joint angle are understood can a muscle's contribution to strength be established.

These data are suitable for use in implementing neuromusculoskeletal models of the glenohumeral joint and upper arm, especially if the models require estimates of optimal muscle length and tendon length. The examination of variability and power increases our understanding of this often used method of predicting optimal muscle length as well as our understanding of the affects of variability in muscle sarcomeres and optimal muscle length on predicted muscle force.

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